Session Objectives

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By the conclusion of the session, the student will be able to:

1. Describe and discuss the etiologies, pathogenesis, pathology, and findings **in benign renal tumors** including **Adenoma, Angiomyolipoma, and Oncocytoma**
2. Describe and discuss the etiology, pathogenesis, pathology, and findings in **Renal Cell Carcinoma**
3. Describe and discuss the etiology, pathogenesis, pathology, and findings in **Wilms tumor**
4. Describe and discuss the etiology, pathogenesis, and findings in **Transitional Carcinoma of the renal pelvis**

Source: Course Syllabus

Kidney Tumors

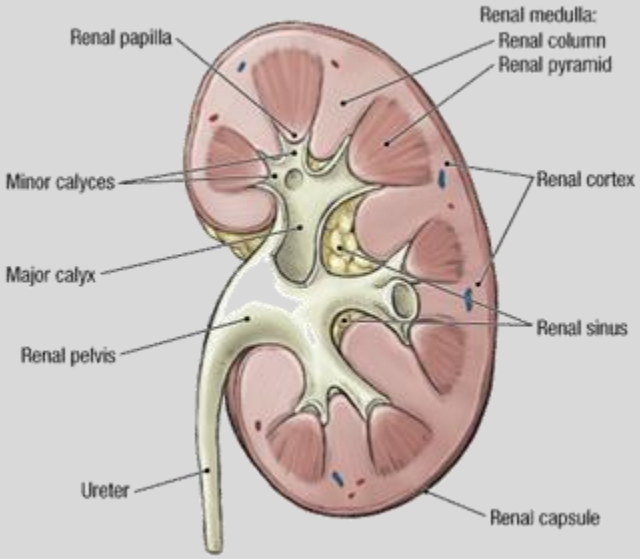
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Benign

- Renal (Papillary) Adenoma
- Angiomyolipoma
- Oncocytoma

Malignant

- Renal Cell Carcinoma
- Urothelial (Transitional Cell) Carcinoma of Renal Pelvis
- Wilms Tumor (mainly pediatric)



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Chapter 20: The Kidney; Grants Dissector, 16e

Adenoma

- Benign tumor of renal tubular epithelium
- Usually papillary: “papillary adenoma”
  - Another rare variety: “metanephric adenoma”
- Small, cortical, well-circumscribed
  - **Usually less than 1.5cm** (varies by source**); size cutoff is >3 cm**
- Clinical:
  - Uncertain malignant potential (treat as carcinoma)
- Gross:
  - Pale yellow-gray, discrete, well-circumscribed nodules
- Histology:
  - Complex, branching, papillomatous structures with numerous complex fronds
  - Cells are cuboidal to polygonal with small, central nuclei, little cytoplasm, no atypia

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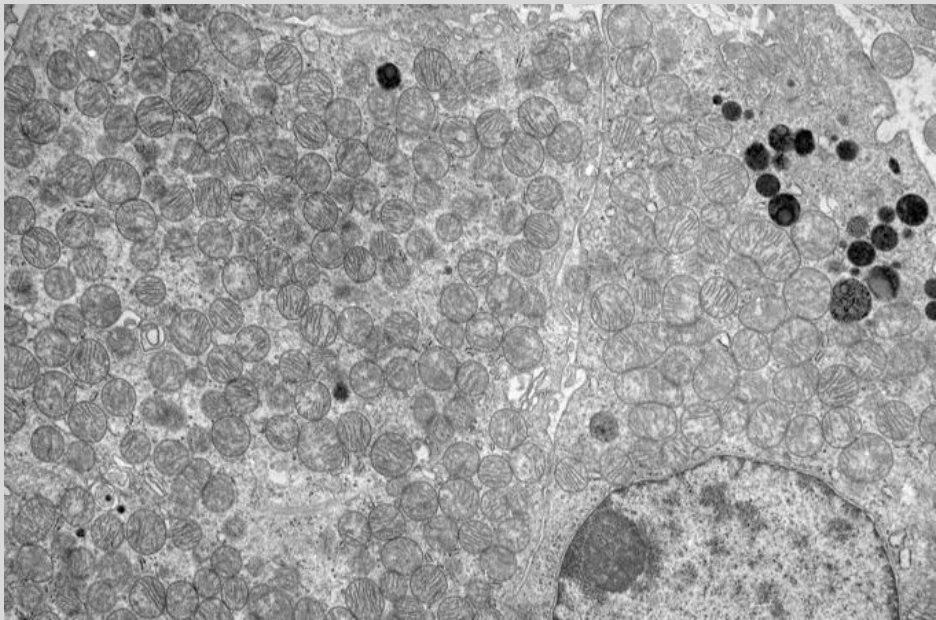
Angiomyolipoma

- **Benign tumor** originating from perivascular epithelioid cells
- Composed of **blood vessels, smooth muscle, and adipose tissue**
- Sporadic (more common in women)
  - Angiomyolipoma cells express estrogen receptor
- Clinical features
  - Mostly asymptomatic
  - Spontaneous retroperitoneal hemorrhage
  - Hematuria
  - Impaired renal function
- Increased frequency in **tuberous sclerosis (TS)**
  - LOF in ***TSC1*** (Hamartin, Chr 9) and ***TSC2*** (Tuberin, Chr 16) tumor suppressor genes
  - Interact to inhibit activity of mammalian target of rapamycin (**mTOR**), a kinase that stimulates cell growth
  - Occur in 25-50% of TS cases

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Oncocytoma

- Benign tumor
  - **Rarely metastasizes**
- 5-10% of resected renal neoplasms
  - Male > female (slightly)
- Thought to arise from **intercalated cells** of the collecting ducts
- Gross:
  - Well-circumscribed, homogenous, encapsulated, **mahogany-brown**
  - **Central scar** may be present
- Histology:
  - Granular, eosinophilic cytoplasm
  - Uniform round nuclei, central nucleoli
  - Many **mitochondria (EM)**



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Renal Cell Carcinoma (RCC)

- **Malignant tumor:** Represents 3% of all newly diagnosed cancers in the U.S. and account for 85% of renal cell cancers in adults; most common primary cancer in kidney
- Mostly **sporadic**; can be inherited
  - 5% are familial - More likely to be bilateral and arise younger (adults)
- Occurs in older individuals (average ~60yrs)
  - Male > female
  - **Tobacco use** is a significant risk factor
  - Additional risk factors: obesity, HTN, unopposed estrogen therapy, asbestos/petroleum/heavy metal exposure
  - Inc. risk in pt. with ESRD, CKD, acquired cystic disease and tuberous sclerosis
- Arises from **renal tubules**

Renal Cell Carcinoma - continued

- **Five subtypes:**
  - **Clear cell (most common)**
  - **Papillary**
  - **Chromophobe**
  - **Xp11 translocation**
  - **Bellini duct aka Collecting duct**
- Tends to invade **renal vein (may extend into IVC)**
- Tends to metastasize widely and early (before s/s)
- 15% of patients have evidence of metastases at time of presentation (most commonly to lungs and bone followed by regional lymph nodes, liver, adrenal glands, and brain)
- 5-year survival = 45%; without metastasis = 70%
- Treatment: partial or radical nephrectomy depending on tumor location/size (<4 cm)/comorbidities (e.g., DM); drugs that inhibit VEGF (e.g., Bevacizumab) and various tyrosine kinases for metastatic

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Renal Cell Carcinoma – Familial

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- 5% Familial/Syndromic variants
  - More likely to be bilateral and arise earlier (younger adults)
- **Von Hippel-Lindau (VHL) syndrome**
  - Renal cysts and bilateral clear cell RCC (**ccRCC**)
  - **Hemangioblastomas** of cerebellum and retina
  - **Pheochromocytoma** (tumor of the adrenal medulla)
  - VHL gene, **AD (3p)**
- **Tuberous Sclerosis**
  - Autosomal Dominant
  - *TSC1* – Hamartin (9q) and *TSC2* – Tuberin (16p)
  - Angiomyolipoma is more common in TS than ccRCC
- **Hereditary Papillary Carcinoma**
  - Autosomal Dominant
  - Mutations in **MET** proto-oncogene
  - Presents with multiple bilateral tumors

*__VHL__ and __TS__ are both associated with **hereditary clear cell carcinoma**

Hereditary RCC Overview

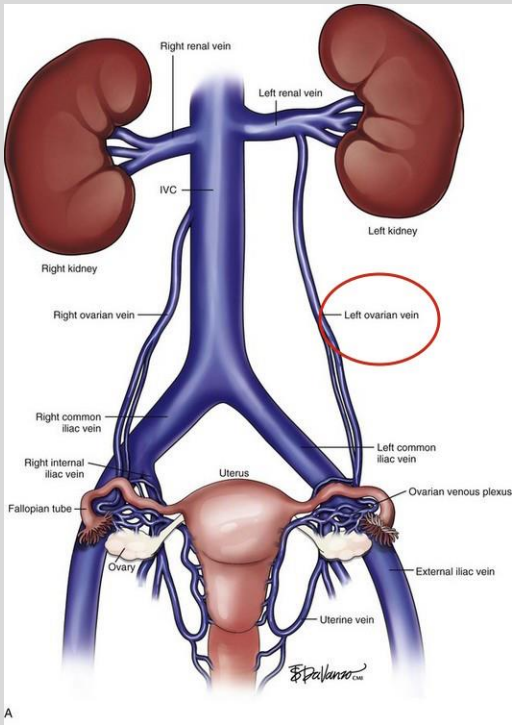
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Syndrome	Gene	Histology	Inheritance Pattern	Other systems Involved
Von Hippel-Lindau	VHL gene (chr. 3p) →overactivation of HIF → angiogenesis, cell growth, cell metabolism	Clear cell carcinoma	Autosomal dominant	Renal cysts and bilateral clear cell RCC (ccRCC) **Hemangioblastomas of cerebellum and retina** **Pheochromocytoma** (tumor of the adrenal medulla)
Tuberous sclerosis	TSC1 (chr. 9) & TSC2 (chr. 16): LOF in TSC1/TSC2 tumor suppressor genes → inhibit mTOR (kinase that stimulates cell growth)	Clear cell carcinoma (also angiomyolipoma which is more common in TS)	Autosomal dominant	Cardiac rhabdomyomas Renal angiomyolipomas Retinal glial hamartomas Cutaneous angiofibromas
Hereditary Leiomyomatosis and renal cell cancer syndromes	FH gene	Papillary	Autosomal dominant	Leiomyomas of skin and uterus PET-positive adrenal adenomas
Birt-Hogg-Dube Syndrome	BHD gene	Chromophobe/hybrid oncocytoma	Autosomal dominant	Cutaneous fibrofilliculoma or trichodiscoma Pulmonary cysts Spontaneous pneumothorax
Hereditary Papillary carcinoma	MET protooncogene AKA hepatocyte growth factor-mediates growth, cell mobility, invasion, and morphogenetic differentiation	Papillary	Autosomal dominant	N/A

Adapted from JNCNN.org & Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney

Renal Cell Carcinoma, Clinical

- **Symptoms:** rarely see all of these
  - **Costovertebral/flank pain** or abdominal pain
  - Palpable flank or abdominal mass
  - **Hematuria** (intermittent/microscopic)
  - Sometimes constitutional symptoms (e.g. fever)
- **Paraneoplastic syndromes ("PEAR"-aneoplastic)**
  - PTHrP: Hypercalcemia
  - Ectopic EPO: Polycythemia
  - ACTH: Cushing syndrome from steroid hormone or ACTH release
  - Renin release: hypertension
- Left-sided varicocele in males (rare)
  - Tumor blocks left renal vein (drainage pathway of left testicular vein)



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Clear Cell RCC (ccRCC)

- **Most common** type of RCC (70-80%)
- Unilateral, more common in upper pole of kidney
- Originates in the **proximal tubule**
- Loss of short arm of **chromosome 3**
  - Sporadic (95%), familial or associated with VHL syndrome
  - VHL tumor suppressor gene is within chromosome 3
  - VHL targets transcription factor hypoxia inducible factor 1 (HIF-1) → elevated HIF-1 → promotion of angiogenesis (VEGF), cell growth (IGF-1), reprograms cell metabolism (MYC)

Histology:

- Round-polygonal cells with clear cytoplasm (due to abundance of lipids/glycogen)
- Non-papillary
- Delicate branching vasculature (**"chicken-wire" pattern**)

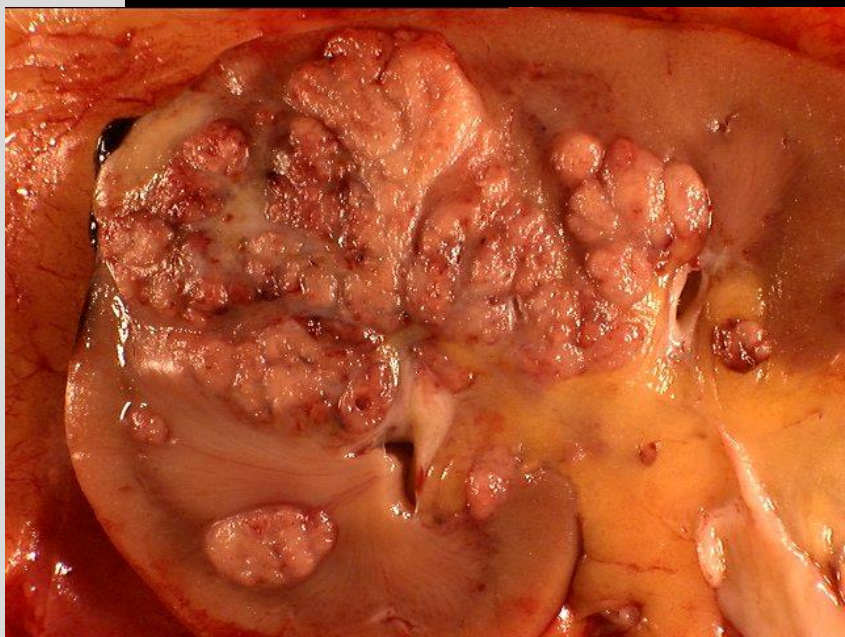
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Papillary RCC

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- 10-15% of RCC cases (less than ccRCC)
- Usually **multifocal**
- Most common tumor in patients who develop **dialysis-associated cystic disease**
- **Multifocal, hemorrhagic,** and **cystic** on gross pathology
- Sporadic:
  - Associated with **Trisomies 7, 16, 17, and loss of Y chr** (males)
- Familial:
  - Associated with **Trisomy 7** (Chr 7 locus for **MET** proto-oncogene)
  - MET: AKA hepatocyte growth factor-mediates growth, cell mobility, invasion, and morphogenetic differentiation





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Chromophobe RCC

- 5% of RCC cases
- Multiple chromosome losses
- Very **good prognosis** (least aggressive)
- Thought to arise from **intercalated cells** (like oncocytomas)
  - This is malignant, oncocytomas are benign
- Histology:
  - Prominent cell membranes
  - **Pale eosinophilic cytoplasm**
    - Often have halo around nucleus
    - **"vegetable cell wall"** appearance
  - Often hard to distinguish microscopically from oncocytomas

Well-circumscribed, globular, solid, tan or brown



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Wilms Tumor (Nephroblastoma)

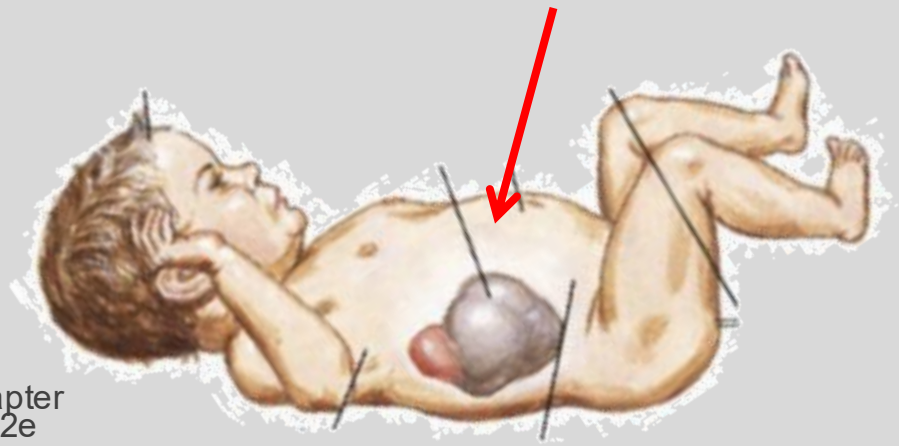
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- ***Most common primary renal tumor of childhood***
  - 4th most common **pediatric** malignancy in the U.S.
  - Peak age: between 2-5 years old
- 3 chromosomal disorders associated with increased risk:
  - **WAGR syndrome:** deletion of **11p13** (***WT1*** gene)
    - **W**ilms tumor (33% chance), **A**niridia, **G**enital anomalies, intellectual disability (formerly mental **r**etardation, inappropriate term: **R**ange of developmental delays)
    - Autosomal dominant
  - **Denys-Drash syndrome:** ***WT1*** gene → 90% risk of Wilms
    - Gonadal **dysgenesis** and early-onset nephropathy/renal failure (**diffuse** mesangial sclerosis)
  - **Beckwith-Wiedemann syndrome (BWS):** deletion of **11p15**
    - Organomegaly, macroglossia, hemihypertrophy, omphalocele, and enlarged adrenals
- 90% of cases are **sporadic** (nonsyndromic) and unilateral tumors
  - 5-10% are bilateral and have a median onset 10 months earlier than unilateral. These patients are assumed to harbor germline mutations.

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Wilms Tumor – Clinical Features

- Large abdominal mass
- Hematuria, hypertension (due to renin release), intestinal obstruction
- DDX: Neuroblastoma
- Pulmonary metastasis can be present at time of diagnosis
  - Other common sites are local nodes and liver
- **Good prognosis** with **chemotherapy and resection**
- Possible association with:
  - Sarcoma
  - Leukemia, lymphoma
  - Brain tumors
  - GU tumors



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Transitional Cell Carcinoma (TCC)

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- aka **Urothelial Carcinoma**
- 90% of bladder tumors, ~5% of renal tumors
  - Renal → appears in **renal pelvis and calyces,** up to distal urethra (covered by transitional epithelium)
- Chromosome **9p deletion** in superficial papillary and noninvasive *in situ* lesions; Chromosome **17p deletions** and **p53 mutations** in invasive TCC
- Male > female, 50-80 years of age
- Risk factors: cigarette smoking, exposure to polycyclic aromatic hydrocarbons or aromatic amines, long-term analgesic use, radiation, family history (first-degree relative)
- Clinical:
  - **Painless** hematuria present in over ¾ cases



**You will learn more about this disease in the 'Pathology: Lower Urinary Tract' lecture*

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Summary Chart

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Tumor	Cell & Location	Clinical & Associations	Gross	Histology
Papillary Adenoma	Renal cortex	Uncertain malignant potential	Small, well-circumscribed	Small central nuclei, little cytoplasm, papillomatous
Angiomyolipoma	Perivascular epithelioid cells	**TS**	Yellowish-tan	Thick-walled vessels, smooth muscle, adipocytes
Oncocytoma	**Intercalated cells**	Benign	Mahogany-brown, Central scar	Homogenous pink cells, many mitochondria
Clear Cell RCC	Proximal tubule	Smoking, VHL, TS; flank pain, hematuria	Colorful, hemorrhage, necrosis, cysts	**"chicken wire" pattern, clear cytoplasm**
Papillary RCC	Distal convoluted tubule	**Trisomy 7; Dialysis-associated cystic disease**	Multifocal, hemorrhagic, and/or cystic	**Foamy macrophages within papillary core**
Chromophobe RCC	**Intercalated cells**	Malignant, good prognosis	Well-circumscribed, solid, tan-brown	**"vegetable cell wall", pale cytoplasm, halo around nucleus**
Collecting Duct RCC	Collecting ducts; renal medulla	Poor prognosis	Like other RCC	Nests of malignant cells within fibrotic stroma
Wilms Tumor	Metanephric blastema	Pediatric; WAGR, DDS, BWS	Large, soft, tan/grey, well-circumscribed, pseudocapsule	**Triphasic (blastema, stroma, epithelioid)**
Transitional Cell Carcinoma	**Transitional epithelium; renal pelvis & calyces**	**Smoking, chemicals; painless hematuria**	Papillary or flat, multifocal	Pleomorphic nuclei, fibrovascular cores

Practice Question 1

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A 58-year-old male presents to the office complaining of blood in his urine. He denies pain when he urinates, but he does experience left costovertebral pain. He has a 20 pack-year history of smoking, but denies using recreational drugs. After a full workup, a biopsy of his left kidney is taken. Histologic analysis shows round-polygonal cells with clear cytoplasm and delicate branching vasculature. What is the most likely site of origin of this mass?

A. Transitional epithelium
B. Proximal tubule
C. Loop of Henle
D. Distal tubule
E. Collecting duct

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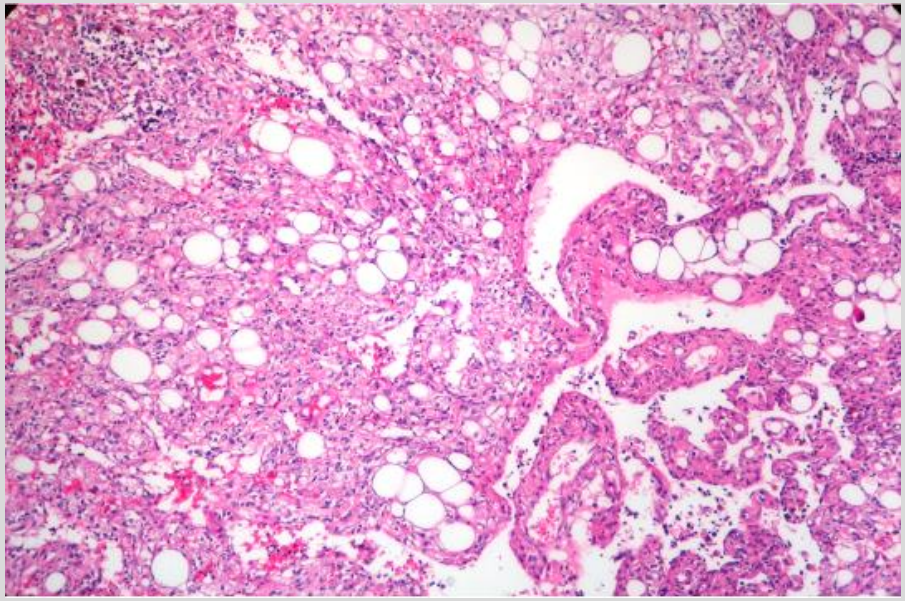
The vignette describes:
Clear cell renal cell carcinoma

Practice Question 2

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A 4-year-old boy presents to the office accompanied by his mother. She explains that she noticed a mass in her son's abdomen that has not gone away with defecation. During the physical exam, areas of cutaneous hypopigmentation are observed on the patient's back as well as multiple erythematous papules on the patient's face. An accompanying biopsy of the patient's abdominal mass is shown. What is the most likely gene mutation associated with this patient's condition?

A. MET
B. NF1
C. VHL
D. TSC1
E. WT1



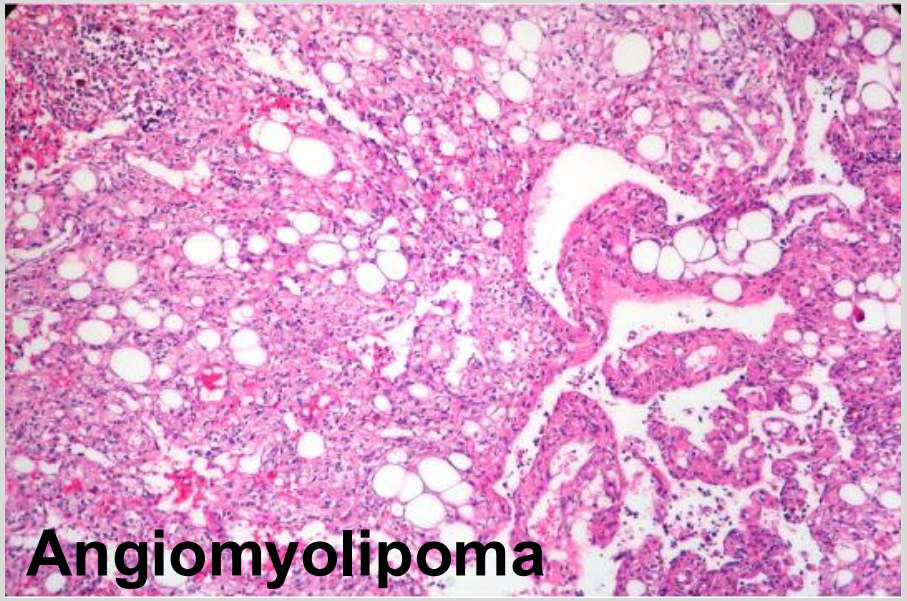
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B. NF1
C. VHL
D. TSC1
E. WT1

The vignette describes:
Tuberous Sclerosis



Practice question 3

A 5-year old has complained of abdominal pain for one month. Vital signs are WNL. Physical exam reveals a palpable mass on the right side. Urinalysis reveals hematuria and proteinuria. Abdominal CT shows a 10 cm circumscribed, solid mass in the right kidney. A right-sided nephrectomy is performed and the following gross appearance of the mass is seen in the figure. What is the most likely diagnosis?

A. Neuroblastoma
B. Angiomyolipoma
C. Renal cell carcinoma
D. Wilms tumor
E. Transitional cell carcinoma



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Practice question 3

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A. Neuroblastoma (may cross midline; constitutional symptoms are common)
B. Angiomyolipoma
C. Renal cell carcinoma
D. Wilms tumor
E. Transitional cell carcinoma



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Summary

- Tumors of the kidney range from benign to highly aggressive malignant.
- Papillary adenomas are benign tumors that tend to be smaller than 1.5 cm.
- Angiomyolipoma is associated with Tuberous Sclerosis.
- Oncocytoma is a (mostly) benign epithelial tumor.
- Renal cell carcinoma is the most prevalent kidney tumor in adults and it tends to metastasize early.
- Some types of renal cell carcinoma include: clear cell, papillary, chromophobe, and collecting duct carcinoma.
- The most common pediatric kidney tumor is Wilms tumor (Nephroblastoma), which is associated with syndromes involving *WT1*.